



## Curriculum Vitae

# Dr. Van Thuy Hoang

## Personal Detail

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| Full name       | Hoang Van Thuy (Mr.)                                                                                                              |
| Date of Birth   | 20-01-1990                                                                                                                        |
| Nationality     | Vietnam                                                                                                                           |
| Current Address | Bucheon city, South Korea                                                                                                         |
| Email           | hoangvanthuy90@gmail.com                                                                                                          |
| Homepage        | <a href="https://hoangvanthuy90.github.io/">https://hoangvanthuy90.github.io/</a>                                                 |
| Google scholar  | <a href="https://scholar.google.com.vn/citations?user=E7Cc8rAAAAAJ">https://scholar.google.com.vn/citations?user=E7Cc8rAAAAAJ</a> |

## Research Interests

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My research focuses on AI for Science & Bioinformatics, analyzing and interpreting biomedical data (molecules). I am interested in using GNNs, Graph Transformers, and Large Language Models to advance knowledge discovery. Key areas of interest include:

- **Foundation Models:** Pre-training & Adaptation strategies, Self-Supervised Learning.
- **Graph Learning:** Graph Structure Learning, Molecular Learning, Knowledge Graphs.
- **AI for Bioinformatics:** Graph Neural Networks, Graph Transformers.
- **Theoretical Foundations:** Information Theory, Causal Inference, Multi-view Learning, Explainable AI, Fairness Learning, Pattern Recognition.

## Education

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| 9/2022 – 3/2026 | Ph.D. in Artificial Intelligence, The Catholic University of Korea, South Korea. |
| 2017 – 2018     | M.S. in Information System, Military Technical Academy, Vietnam.                 |
| 2008 – 2013     | B.S. in Information Technology, Military Technical Academy, Vietnam.             |

## Selected Publications (*Top-tier conferences/journals*)

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- Van Thuy Hoang, O-Joun Lee. *Pre-training Graph Neural Networks on Molecules by using Subgraph-conditioned Graph Information Bottleneck*. In Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI 2025) (**Ranking A\***)
- Van Thuy Hoang, Hyeon-Ju Jeon, O-Joun Lee. *Mitigating Degree Bias in Graph Representation Learning with Learnable Structural Augmentation and Structural Self-attention*. IEEE Transactions on Network Science and Engineering (IEEE TNSE 2025) (**Top 1.8% JCR**)
- Van Thuy Hoang, O-Joun Lee. *Transitivity-Preserving Graph Representation Learning for Bridging Local Connectivity and Role-based Similarity*. In Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI 2024) (**Ranking A\***)
- Nam-Gyu Jung, Van Thuy Hoang, O-Joun Lee, Chang Choi. *Kiosk Recommend System based on Self-Supervised Representation Learning of User Behaviors in Offline Retail*. IEEE Internet of Things Journal, February 2024. (**Top 3.4% JCR**)

## Full Publications

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### 2026:

1. Van Thuy Hoang, O-Joun Lee: Context-aware Graph Causality Inference for Few-Shot Molecular Property Prediction. arXiv preprint arXiv:2601.11135. *Under review*.
2. Van Thuy Hoang, O-Joun Lee: Pre-training Graph Neural Networks on 2D and 3D Molecular Structures by using Multi-View Conditional Information Bottleneck. arXiv preprint arXiv:2511.18404. *Under review*.

### 2025:

3. Van Thuy Hoang, O-Joun Lee. Pre-training Graph Neural Networks on Molecules by using Subgraph-conditioned Graph Information Bottleneck. In Proceedings of the 39th AAAI Conference on Artificial Intelligence 2025 (**AAAI 2025**).
4. Van Thuy Hoang, Hyeon-Ju Jeon, O-Joun Lee. Mitigating Degree Bias in Graph Representation Learning with Learnable Structural Augmentation and Structural Self-attention. IEEE Transactions on Network Science & Engineering 2025. (**Top 1.8% JCR**)
5. Van Thuy Hoang, Tien-Bach-Thanh Do, Jinho Seo, Seung Charlie Kim, Luong Vuong Nguyen, Duong Nguyen Minh Huy, Hyeon-Ju Jeon, O-Joun Lee. Halal or Not: Knowledge Graph Completion for Predicting Cultural Appropriateness of Daily Products. IEEE Access, 2025.
6. Thien Thi Thu Nguyen, Eun-Soon You, Hyun Woo Kim, Van Thuy Hoang, Luong Vuong Nguyen, O-Joun Lee. Constructing a Companion Animal Disease Knowledge Graph by Utilizing LLMs in Data Preprocessing and Pseudo Annotation. Proceedings of the International Conference on Research in Adaptive and Convergent Systems (RACS 2025)

### 2024:

7. Van Thuy Hoang, O-Joun Lee. Transitivity-Preserving Graph Representation Learning for Bridging Local Connectivity and Role-based Similarity. In Proceedings of the 38th AAAI Conference on Artificial Intelligence 2024 (**AAAI 2024**)
8. Nam-Gyu Jung, Van Thuy Hoang, O-Joun Lee, Chang Choi. Kiosk Recommend System based on Self-Supervised Representation Learning of User Behaviors in Offline Retail. IEEE Internet of Things Journal, February 2024. (**Top 3.4% JCR**)
9. Van Thuy Hoang, O-Joun Lee. An Overview of Pre-Trained Graph Models for Molecular Structure Analysis. In IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia 2024).

### 2023:

10. Van Thuy Hoang, T. S. Nguyen, S. Lee, J. Lee, L. V. Nguyen, O.-J. Lee. Companion Animal Disease Diagnostics Based on Literal-Aware Medical Knowledge Graph Representation Learning. IEEE Access, 2023.
11. Van Thuy Hoang, Hyeon-Ju Jeon, Eun-Soon You, Y.W. Yoon, Sungyeop Jung, O-Joun Lee. Graph Representation Learning and Its Applications: A Survey. Sensors, 2023.

### 2018:

12. Van Thuy Hoang, Phan Viet Anh, and Nguyen Xuan Hoai. "Automated large program repair based on big code." In Proceedings of the 9th International Symposium on Information and Communication Technology 2018 (ISICT 2018)
13. Ly Vu, Van Thuy Hoang, Uy Nguyen, Tran Ngoc, Diep Nguyen, Hoang Dinh, Eryk Dutkiewicz. Time Series Analysis for Encrypted Traffic Classification: A Deep Learning Approach. In Proceedings of the 18th International Symposium on Communications and Information Technologies 2018 (ISCIT 2018).

### 2013:

14. Loi Cao Van, Van Thuy Hoang, Quang Uy Nguyen. A scheme for building a dataset for intrusion detection systems. 2013 Third World Congress on Information and Communication Technologies (WICT 2013)

## Patents

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1. O-Joun Lee, Van Thuy Hoang: System for predicting molecular property and method thereof (분자 특성 예측 시스템 및 그 방법). Ref. No: KR 1020260008863, 01/2026
2. O-Joun Lee, Van Thuy Hoang: Method for pre-training graph neural network for 2D and 3D molecular structures and system thereof (분자의 2차원 및 3차원 구조에 대한 그래프 신경망 사전학습 방법 및 그 시스템). Ref. No: KR 1020250169745, 11/2025
3. O-Joun Lee, Van Thuy Hoang: Method for pre-training using graph information bottleneck based on subgraph and system thereof (서브그래프 기반 그래프 정보 병목 기법을 이용한 사전학습 방법 및 그 시스템). Ref. No: KR 1020250030478, 03/2025
4. O-Joun Lee, Van Thuy Hoang, Hochan Yang, Ju Hee Shim, Seung Kim, SungGyu Kim: System of Recommending Halal Cosmetics based on Knowledge Graph Representation Learning (지식 그래프 표현 학습에 기반한 할랄 화장품 추천 시스템). Ref. No: KR 1020240159542, 11/2024
5. O-Joun Lee, Van Thuy Hoang: Graph Representation Learning Method and System Thereof (그래프 표현 학습 방법 및 그 시스템). Ref. No: KR 1020240104783, 08/2024
6. O-Joun Lee, Van Thuy Hoang, Sang Thanh Nguyen, Dong Kook Park, Doyul Lee: Apparatus and Method for Representation Learning of Knowledge Graph using Heterogeneous Data Comprehensively (이종 데이터를 통합적으로 이용하는 지식 그래프 표현 학습 방법 및 장치). Ref. No: KR 1020240090256, 07/2024
7. O-Joun Lee, Van Thuy Hoang: Apparatus and Method for Training Graph Neural Network based on Graph Augmentation (그래프 증강 기반 그래프 뉴럴 네트워크 학습 방법 및 장치). Ref. No: KR 1020240086072, 07/2024
8. O-Joun Lee, Van Thuy Hoang, Ho Beom Kim, Jinho Seo, Seung Charlie Kim, Nox Kim: System and Method for Predicting Availability based on Knowledge Graph (지식 그래프 기반의 가용성 예측 시스템 및 방법). Ref. No: KR 1020230171167, 11/2023

## Academic Services

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- Conference Reviewer: AAAI-26, ICLR-26
- Journal Reviewer: Pattern Recognition, Journal of Big Data, Neurocomputing, Knowledge-Based Systems, Expert Systems with Applications, npj Computational Materials, Information Sciences.

## Preprint

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15. Van Thuy Hoang, O-Joun Lee: A Survey on Structure-Preserving Graph Transformers. 01/2024. arXiv preprint. arXiv:2401.16176.
16. Thanh Sang Nguyen, Jooho Lee, Van Thuy Hoang, O-Joun Lee: Connector 0.5: A unified framework for graph representation learning. 04/2023. arXiv preprint. arXiv:2304.13195

## References

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**Prof. O-Joun Lee**

Assistant Professor

The Catholic University of Korea (CUK), South Korea.

Ph.D. Supervisor                      E-mail: ojlee@catholic.ac.kr

**Prof. Nguyen Xuan Hoai**

Associate Professor

Co-founder at AI Academy Vietnam; Assoc. Prof., Ho Chi Minh City University of Technology (HUTECH), Vietnam.

M.S. Supervisor                      E-mail: nxhoai@aiacademy.edu.vn

**Dr. Cao Van Loi**

Military Technical Academy, Vietnam.

B.S. Supervisor                      E-mail: loi.cao@lqdtu.edu.vn